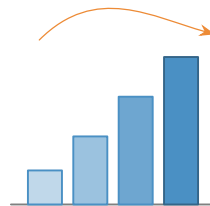


Chapter 1

Ten Principles of Economics

This chapter previews the ten big ideas that the rest of the course keeps coming back to.

We ask three kinds of questions: how individual people make decisions, how people interact with one another, and how the economy as a whole works.



What Is Economics?

A household must decide how to allocate its members' time, money, and effort among competing needs.

A society faces exactly the same problem, only on a much larger scale: it must decide who does which job, and who gets to consume the goods and services that result.

Definition. **Scarcity** is the limited nature of society's resources.

Definition. **Economics** is the study of how society manages its scarce resources.

How Economists Study the World

Because resources are scarce, society cannot produce everything that everyone wants, so economists study three things.

- Economists study how individual people make decisions: how much they work, what they buy, how much they save, and how they invest their savings.
- Economists study how people interact with one another, for example how buyers and sellers together determine the price and quantity of a good.
- Economists study the forces and trends that affect the economy as a whole, including the growth of average income, the fraction of people who cannot find work, and the rate at which prices rise.

How People Make Decisions

The first four principles concern individual decision making, since the behavior of an economy simply reflects the behavior of the individuals who make it up.

- Principle 1: People face trade-offs.
- Principle 2: The cost of something is what you give up to get it.
- Principle 3: Rational people think at the margin.
- Principle 4: People respond to incentives.



Principle 1: People Face Trade-Offs

There is no such thing as a free lunch: to get one thing we like, we usually have to give up another. Making a decision means trading off one goal against another.

- A student who spends an evening studying economics gives up an evening she could have spent studying psychology, or napping, or working at a part-time job.
- A society that spends more on national defense has less to spend on consumer goods; a society that raises pollution regulations to protect the environment raises the cost of producing goods and services.

Example: Taiwan's Defense Budget Trade-Off

Taiwan faces this trade-off directly: as cross-strait tensions push defense spending toward a larger share of GDP, that same budget cannot also fund national health insurance, pensions, or public housing.

Any honest debate over the defense budget is really a debate about what else Taiwan gives up to pay for it.

In recent years, Taiwan's government has pushed to raise defense spending toward, and in some proposals beyond, 3 percent of GDP, a level that would once have seemed politically unlikely.

The same logic runs in reverse: a decision to expand national health insurance coverage or raise pension benefits is just as much a decision to leave less room in the budget for defense.

Trade-Offs Society Faces: Efficiency versus Equality

Definition. **Efficiency** is the property of society getting the most it can from its scarce resources.

Definition. **Equality** is the property of distributing economic prosperity uniformly among the members of society.

Policies that redistribute income from the rich to the poor can make society more equal. But they also reduce the reward for working hard, so they shrink the total size of the economic pie even as they slice it more evenly.

Recognizing this trade-off does not tell us what a society should choose. It does mean, however, that good policy starts from acknowledging that the trade-off exists.

Principle 2: The Cost of Something Is What You Give Up to Get It

Because people face trade-offs, making a decision requires comparing the costs and benefits of alternative courses of action. Often, though, the true cost of an action is not as obvious as it first appears.

Definition. The **opportunity cost** of an item is whatever must be given up to obtain that item.

Example. The opportunity cost of a year in college is not simply tuition, books, and fees. Room and board are not really extra costs, since you would have paid for food and a place to sleep either way.

What you should add instead is the income you gave up by not working during that year. For most students, that forgone income is the single largest cost of their education.

Example: The Opportunity Cost of Military Service

Most Taiwanese men must complete a period of mandatory military service.

Conscript pay during that period is far below what most could earn in a civilian job. That gap between conscript pay and civilian pay, not the pay itself, is the opportunity cost of service.

That is why some conscripts choose alternative civilian service when it is available.

It is also why the policy debate over how long service should last is really a debate about how large an opportunity cost the country asks young men to bear.

Principle 3: Rational People Think at the Margin

Definition. **Rational people** are people who systematically and purposefully do the best they can to achieve their objectives, given the opportunities available to them.

Definition. A **marginal change** is a small incremental adjustment to an existing plan of action.

A rational decision maker takes an action if, and only if, the marginal benefit of that action exceeds its marginal cost.

This is why an airline is willing to sell a standby seat for far less than the average cost per passenger. Once the plane is going to fly anyway, the marginal cost of one more passenger is tiny.

Marginal Thinking: The Water–Diamond Paradox

Water is essential to life, while diamonds are not. Yet people are willing to pay far more for a diamond than for a cup of water.

Marginal thinking resolves this puzzle. Water is plentiful, so the marginal benefit of one more cup is small. Diamonds are rare, so the marginal benefit of one more diamond is large.

The same logic explains why a streaming service can set the marginal cost of watching one more movie to zero.

Once you have paid the monthly fee, watching another film costs the company almost nothing extra, so it is willing to let you watch as much as you like.

Example: When Taiwan Ran Short of Water

The water-diamond paradox is not just a textbook curiosity. During Taiwan's severe drought in 2021, reservoirs in central Taiwan fell so low that the government cut irrigation to farmland, to protect water for households and semiconductor fabs such as TSMC, which need enormous, reliable water supplies to make chips.

Water did not stop being essential during the drought, but it did stop being plentiful, so its marginal value rose sharply. The economy responded exactly as marginal thinking predicts:

- Farmers were compensated for fallowed fields.
- Factories trucked in water and drilled new wells.
- The price the economy was effectively willing to pay for one more liter of water shot up.

Discussion: Thinking at the Margin

Consider the following two situations. In each case, identify the marginal benefit and marginal cost, and decide what a rational decision maker should do.

- As the manager of a local grocery store, you are considering hiring one more cashier. The new hire would raise sales revenue by NT\$12,000 per week but would earn NT\$9,000 per week in wages. Should you hire the cashier?
- You pay a flat NT\$150 per month for a streaming subscription, no matter how much you watch. Should you watch one more episode tonight?

Answer: Thinking at the Margin

In the grocery store example, the marginal benefit of hiring the cashier is the NT\$12,000 increase in weekly sales, and the marginal cost is the NT\$9,000 weekly wage. Because the marginal benefit exceeds the marginal cost, the manager should hire the additional cashier.

In the streaming example, the monthly fee is already paid and does not change with how much you watch. So the marginal monetary cost of one more episode is zero.

The only real marginal cost is the opportunity cost of the time you spend watching. As long as the enjoyment you get from the episode exceeds that opportunity cost, you should watch it.

Principle 4: People Respond to Incentives

Definition. An **incentive** is something that induces a person to act, such as the prospect of a punishment or a reward.

Because rational people make decisions by weighing costs and benefits, they respond when those costs and benefits change.

When the price of doughnuts rises, the higher price works as two opposite signals:

- It is a disincentive to buyers, who buy fewer doughnuts.
- It is an incentive to sellers, who produce more.

Policymakers should never forget this: any policy that changes the costs or benefits people face will also change their behavior, sometimes in ways the policy never intended.

Incentives Working as Intended: Taipei's Trash Policy

Before the late 1990s, Taipei residents paid a flat fee for garbage collection no matter how much they threw away. Throwing away one more bag cost nothing extra.

The city then switched to charging households by the official garbage bag. Every extra bag now has a real marginal cost.

Households responded exactly as the theory predicts. They sorted more recycling and food waste out of the trash. Taipei's household garbage volume fell sharply in the years after the policy began.

Changing the incentive, not changing anyone's values, is what changed the behavior.

Incentives Can Backfire: The Seat Belt Law

When seat belts became standard equipment, they made accidents less costly by reducing the risk of injury. That is exactly what lawmakers intended.

But safer driving also has an opportunity cost: the time and effort it takes to drive slowly and carefully. Seat belts lowered the benefit of driving that way.

The economist Sam Peltzman argued that drivers responded to this changed incentive by driving faster and less carefully. Seat belt laws, he argued, led to more accidents overall, even though they reduced the chance of dying in any single accident.

The lesson is that a policy's indirect effects, working through incentives, can be just as important as its direct effects.

Discussion: Applying the Principles

You are selling your car. You have already spent NT\$60,000 on repairs, and now the transmission has died.

You can pay NT\$42,000 to fix it, or sell the car as is.

1. Suppose the car is worth NT\$430,000 with a working transmission and NT\$330,000 without one. Should you pay to fix it?
2. Suppose instead the car is worth NT\$370,000 with a working transmission and NT\$330,000 without one. Should you pay to fix it?

In each case, compare the *benefit* of the repair with its NT\$42,000 cost.

Notice that the NT\$60,000 you already spent should play no role in the decision.

Answer: Applying the Principles

The NT\$60,000 already spent on repairs is a sunk cost: it is gone regardless of what you decide next.

A rational decision maker should ignore it entirely, and compare only the NT\$42,000 cost of fixing the transmission with the benefit of doing so.

Compare the two scenarios:

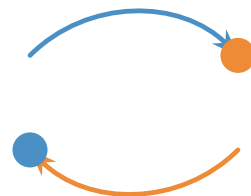
1. In the first, fixing the transmission raises the car's value from NT\$330,000 to NT\$430,000, a benefit of NT\$100,000. That exceeds the NT\$42,000 cost, so you should get it fixed.
2. In the second, fixing it raises the value from NT\$330,000 to NT\$370,000, a benefit of only NT\$40,000. That is less than the NT\$42,000 cost, so you should sell the car as is.

How People Interact

The first four principles concerned individual decisions.

Many of our decisions, however, affect other people as well, so the next three principles concern how people interact with one another.

- Principle 5: Trade can make everyone better off.
- Principle 6: Markets are usually a good way to organize economic activity.
- Principle 7: Governments can sometimes improve market outcomes.



Principle 5: Trade Can Make Everyone Better Off

It is easy to think of trade as a contest in which one side wins and the other loses. That is not how trade works: trade between two countries, like trade between two families, can make both sides better off.

If your family tried to produce everything it consumes by itself, it would have to grow its own food, sew its own clothes, and build its own home.

Trading with others lets each person specialize in what they do best, so that everyone can buy a greater variety of goods and services at a lower cost than they could achieve alone.

Example: Trading Chips for What Taiwan Lacks

Taiwan has almost no domestic oil, coal, or natural gas, yet it imports nearly all the energy and raw materials its economy needs.

It can afford to because it specializes instead in advanced semiconductor manufacturing, exporting chips that the rest of the world is willing to pay a great deal for.

Trade lets Taiwan enjoy a standard of living that self-sufficiency in energy and raw materials could never have supported on its own.

Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

Definition. A **market economy** is an economy that allocates resources through the decentralized decisions of many firms and households as they interact in markets for goods and services.

Centrally planned economies in the Soviet Union and Eastern Europe collapsed because planners lacked the information needed to decide what to produce, how, and for whom.

In a market economy, no single person is looking out for the well-being of society as a whole. And yet market economies have proven remarkably successful at promoting overall prosperity.

Example: Markets at Work on Every Corner

Taiwan is often cited as having one of the highest densities of convenience stores per person anywhere in the world, with chains such as 7-Eleven and FamilyMart competing block by block.

No planning agency decided how many stores each neighborhood needed.

Each store's owner responded to the same signal: the profit to be made from serving nearby customers.

Their combined decisions produced a distribution network that is remarkably convenient, without anyone designing it that way. That is the decentralized coordination Principle 6 describes.

Adam Smith's Invisible Hand

Prices are the instrument through which a market coordinates the decisions of buyers and sellers. Buyers look at the price to decide how much to demand. Sellers look at the price to decide how much to supply.

Market prices therefore come to reflect both the value of a good to buyers and the cost of producing it.

In 1776, the economist Adam Smith observed that households and firms interacting in markets behave as though they are guided by an invisible hand. That hand steers them toward outcomes that, in many cases, maximize the well-being of society as a whole.

When a government prevents prices from adjusting freely, whether through a tax or through price controls, it ties the invisible hand behind its back. It distorts the decisions that households and firms would otherwise make.

Principle 7: Governments Can Sometimes Improve Market Outcomes

The invisible hand can only work if the government enforces the rules a market economy depends on.

Definition. Property rights are the ability of an individual to own and exercise control over scarce resources.

Without secure property rights, people stop producing:

- A farmer will not grow food if she expects the crop to be stolen before harvest.
- A film studio will not produce movies if it cannot stop people from copying them for free.

The invisible hand depends on government enforcement of these rights.

Government and Market Failure

Even when property rights are secure, the invisible hand is not all-powerful.

Definition. A **market failure** is a situation in which a market left on its own fails to allocate resources efficiently.

Market failure has two common sources:

- An **externality**: the effect of one person's actions on a bystander, such as pollution.
- **Market power**: a single actor's ability to meaningfully influence market prices, as in a monopoly.

Well-designed policy can improve efficiency in either case. Government can also use tax and welfare policy to pursue greater equality. But that government *can* improve outcomes does not mean it always will.

Example: Taiwan's National Health Insurance

Left to itself, a private health insurance market tends to fail. Insurers want to avoid the sickest customers, while healthy people are tempted to skip coverage until they need it. So the market can unravel toward too little insurance for the people who need it most.

Taiwan's single-payer National Health Insurance program addresses this by making enrollment mandatory for essentially everyone.

This single policy serves both of the goals Principle 7 describes:

- It corrects a market failure, since near-universal enrollment prevents the market from unraveling.
- It promotes equality, since access to care no longer depends primarily on a person's income or employer.

Discussion: The Government's Role

For each situation below, ask: What is the government's role? Is the intervention meant to promote efficiency or equality? Does it actually improve the outcome?

- Nine years of compulsory education for every child.
- Workplace safety regulations.
- Taiwan High Speed Rail, built and operated under a government-granted concession to a private company.
- Patent laws that let drug companies charge high prices for life-saving drugs.

How the Economy as a Whole Works

Individual decisions and the interactions among people together make up what we call “the economy.”

The last three principles concern the workings of the economy as a whole.

- Principle 8: A country’s standard of living depends on its ability to produce goods and services.
- Principle 9: Prices rise when the government prints too much money.
- Principle 10: Society faces a short-run trade-off between inflation and unemployment.

Principle 8: Standard of Living and Productivity

Living standards vary enormously both across countries and over time:

- Average income today is many times higher in the richest countries than in the poorest.
- Average income in the United States has roughly doubled every 35 years at its historical growth rate.

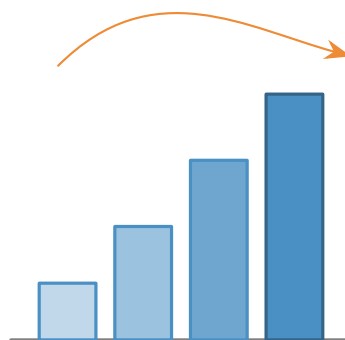
Definition. Productivity is the quantity of goods and services produced from each unit of labor input.

Almost all of this variation traces back to differences in productivity, not to commonly blamed causes such as labor unions or foreign competition. Raising living standards means, first and foremost, raising productivity.

Example: The Taiwan Miracle

Taiwan's own history illustrates the point. In just a few decades, Taiwan moved from a mostly agricultural economy to a manufacturing base and then into advanced semiconductor production, a rise often called the “Taiwan Miracle.”

Average income rose accordingly, not because Taiwanese workers began working more hours, but because each hour of their work came to produce far more value.



The Taiwan Miracle, Continued

This did not happen by accident. Postwar land reform, export-oriented industrial policy, and heavy investment in education all raised the productivity of Taiwanese workers, sector by sector.

Taiwan is often grouped with South Korea, Hong Kong, and Singapore as one of the “Four Asian Tigers,” economies that grew from among the world’s poorest to among its richest within a single generation.

The clearest modern example is semiconductor manufacturing. Firms such as TSMC now produce chips the rest of the world depends on, generating a value added per worker that would have been unimaginable when Taiwan was still a mostly agricultural economy.

Principle 9: Prices Rise When the Government Prints Too Much Money

Definition. Inflation is an increase in the overall level of prices in the economy.

In Germany in the early 1920s, prices rose so fast that a newspaper costing 0.30 marks in January 1921 cost 70,000,000 marks by November 1922, less than two years later.

In almost every case of high or persistent inflation, the underlying cause is the same: the government has created money faster than the economy is producing goods and services. Each unit of money then comes to be worth less.

The faster a government prints money, the higher the resulting rate of inflation.

Example: Taiwan's Own Hyperinflation

Taiwan does not need to look only to Weimar Germany for a cautionary tale.

In the years after 1945, wartime and postwar money creation drove prices in Taiwan up so far that, in 1949, the government retired the old Taiwan Dollar. It replaced it with a New Taiwan Dollar at a conversion rate on the order of tens of thousands of old dollars to one new dollar.

That reform is the reason Taiwan's currency today is called the New Taiwan Dollar.

It is a domestic reminder that the link between excessive money creation and runaway prices is not just a European or historical curiosity.

Principle 10: The Short-Run Trade-Off between Inflation and Unemployment

Although money growth determines inflation in the long run, the short-run story is more complicated.

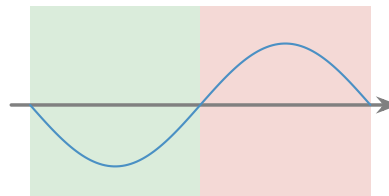
An increase in the amount of money in the economy stimulates spending, which encourages firms to raise prices only gradually while hiring more workers and producing more in the meantime. More hiring means lower unemployment.

This chain of reasoning leads to one further trade-off: over a period of a year or two, many economic policies push inflation and unemployment in opposite directions.

Economists still debate exactly how policymakers should use this short-run trade-off, but almost none doubt that it exists.

Definition. The **business cycle** refers to fluctuations in economic activity, such as employment and production.

This trade-off is central to the business cycle: a policymaker fighting a recession by expanding the money supply is, in effect, accepting higher inflation for lower unemployment.



Discussion: A Cheaper Scooter Permit

Suppose NTU cuts the price of a campus scooter parking permit from NT\$3,000 per semester to NT\$100 per semester. Think through the following questions, keeping the idea of opportunity cost in mind.

- What happens to the number of students who want to park a scooter on campus?
- What happens to the amount of time it takes to find a parking spot?
- Does the lower price of the permit necessarily lower the true cost of parking?
- Would parking's opportunity cost be the same for a student with no job as for a student earning NT\$200 an hour?

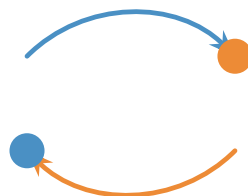
Chapter in a Nutshell: Individual Decisions

- People face trade-offs among alternative goals.
- The true cost of any action is measured in terms of the opportunities it forces us to give up.
- Rational people make decisions by comparing marginal benefits with marginal costs.
- Everyone changes their behavior in response to the incentives they face.



Chapter in a Nutshell: How People Interact

- Trade and interdependence between people, and between countries, can make everyone involved better off.
- Markets are usually a good way of coordinating economic activity.
- Governments can potentially improve on what markets achieve on their own, either by correcting a market failure or by promoting greater equality.



Chapter in a Nutshell: The Economy as a Whole

- Productivity is the ultimate source of a country's living standard.
- Growth in the quantity of money is the ultimate source of inflation.
- In the short run, society faces a trade-off between inflation and unemployment, even though that trade-off eventually fades as prices adjust.

These ten principles are only a preview.

We do not expect them to be fully convincing yet, since we will spend the rest of this course exploring each one in far greater depth. But every idea we cover from here on builds on one or more of them.

Key Concepts

- Scarcity, economics, efficiency, and equality describe the resource problem that all of economics studies.
- Opportunity cost, rational people, marginal change, and incentive describe how an individual decision maker weighs a choice.
- Market economy, property rights, market failure, externality, and market power describe how people interact and where markets can fall short.
- Productivity, inflation, and the business cycle describe the workings of the economy as a whole.